widely scattered from Cader Idris to Conway, have all been rotated by the same amounts by a bodily movement of North Wales, then their *in situ* remanences must be considered to define the geomagnetic field at the time when those remanences were acquired.

In North Wales there is no geological evidence of any large scale post-Caledonian thermal event at high enough temperatures to explain the stable remanent magnetisations observed in this study. In any case the only known palaeomagnetic field corresponding to these results is that of Tertiary to Recent times. The small amount of Tertiary igneous activity in North Wales could not have been responsible for magnetising rocks on the large scale necessary to explain the observations.

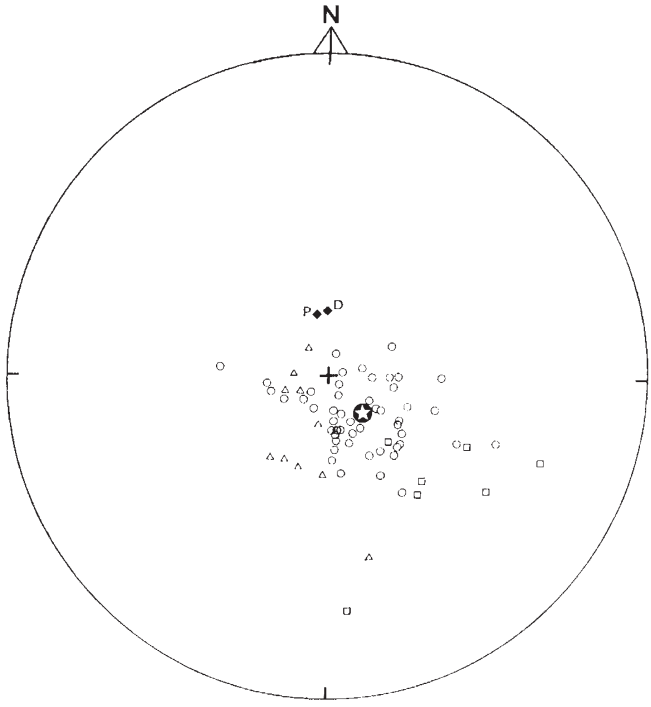


Fig. 1 Stereographic projection of site mean remanences after a.f. cleaning of intrusives from North Wales, and the basic lavas of Cader Idris. ○, *In situ* remanences in all intrusive rocks except Myndd Nefyn; the star indicates their overall mean direction; □, the *in situ* results from Myndd Nefyn; △, tilt-adjusted results from the basic lavas of Cader Idris. The present axial geocentric dipole and the present geomagnetic field directions are marked ◆ and labelled D and P respectively. Open symbols indicate negative (upwards) inclinations and closed symbols positive (downwards) inclinations.

All conventional interpretations of the results of this study are therefore unsatisfactory, and instead, the possibility of an anomalous geomagnetic field lasting several million years during the Lower Palaeozoic must be seriously considered. It is most likely that we have recorded only one such anomalous event in this study because of the improbability of successive intrusions repeatedly occurring at similar geomagnetically anomalous instants. Evidence of precisely when this anomaly might have existed is inconclusive. Its occurrence in late Ordovician times would, however, be compatible with the inferred primary TRM of the intrusions, their possible Ordovician age and the similarity of their remanence directions with that of the proven Ordovician Cader Idris lavas.

The reality of an anomalous magnetic field lasting $\sim 10^7$ yr would be of considerable geomagnetic interest. Whether it exists or not should be capable of resolution by palaeomagnetic studies elsewhere in the world.

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Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland

Palaeomagnetic results from a high grade Precambrian terrain in western Greenland show many features attributable to magnetisation during very slow cooling caused by gradual uplift and erosion. They may locate a whole segment of the Laurentian apparent polar wander (APW) path with unusually high precision and also indicate the probable direction of movement.

The magnetic effects described here are analogous to those experienced by radiometric systems during slow cooling[1,2], and are attributable in part to the fact that magnetic minerals have a range of blocking temperatures at which the magnetisation becomes effectively fixed. The effective blocking temperatures of magnetic grains will be considerably lowered during slow cooling, possibly to as low as 200 °C (refs 4 and 5) so that magnetisation would be acquired during the later stages of unroofing, when moderate erosion rates can be expected. Average present-day erosion rates have been variously estimated at 30 and 76 m Myr^{-1} (refs 6 and 7, respectively), and if a slightly higher rate of 100 m Myr^{-1} is assumed together with a geothermal gradient of 20 °C km^{-1}, and a blocking temperature range of 50 °C for stably magnetised grains, then the magnetisation of any one specimen would take 25 Myr. Also, because the magnetisation will be acquired as the rocks pass through the (presumably horizontal) isotherms corresponding to their critical range of blocking temperatures, rocks at higher structural levels will become magnetised earlier than those at lower levels. For example, with an erosion rate of 100 m Myr^{-1}, a difference in level of 2 km would correspond to a difference in the mean time of magnetisation of the order of 20 Myr.

Slow cooling during unroofing should therefore produce several effects in the palaeomagnetic record.

- Apart from the effects of APW all of the rocks formed before uplift should be magnetised in the same direction and give the same pole position, even though geological evidence indicates that they are of different ages.
- Because secular variation should effectively be averaged out within each specimen, between-site scatter attributable to this cause should be small.
- If APW occurred during the magnetisation period then poles from sampling sites at different original structural levels should be strung out along a trend corresponding to that particular section of the APW path, with higher sites at the older part of the trend and lower sites at the younger part.

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● If they have the same mean blocking temperature different sampling sites at the same structural level should give poles close together. Any variation in mean blocking temperature, however, will clearly affect the time of magnetisation, and so poles from sites with high blocking temperatures should be displaced towards the older part of the trend compared with sites with lower blocking temperatures at the same level. Because of this it is unlikely that there will be a perfect relationship between the position of a pole in the trend of poles and the structural level of the corresponding sampling site. Determination of mean blocking temperatures in the laboratory should confirm the sense of time along the trend indicated by comparing the position of poles from sites at different structural levels.

● As any specimen has probably taken millions of years to cool through its critical range of blocking temperatures it should contain the magnetic record of that interval of time. Progressive thermal demagnetisation should therefore effectively move the magnetisation of a specimen from the lowest to the highest blocking temperature direction. Two possibilities arise. First, if the blocking temperature spectrum has two peaks corresponding, for example, to haematite and magnetite, then two distinct magnetisations may be present, an older one corresponding to haematite and a younger one corresponding to magnetite. Progressive demagnetisation would cause the magnetisation to move towards the older direction, but the actual movement would simply follow the great circle connecting the two directions. Such results have been reported from the Grenville province[3,4]. Secondly, if there is a continuous spectrum of blocking temperatures attributable, for example, to a continuous variation in grain size or ionic substitution, then a continuous spread of directions of magnetisation should exist between those corresponding to the highest and lowest blocking temperatures. Progressive demagnetisation would again cause the magnetisation to move towards the older direction, and the actual movement should indicate, after suitable vector analysis, the form of that particular section of the APW path.

● If rocks cool through their critical range of blocking temperatures during a period when the Earth's field is reversing itself at the same rate as at present they would be effectively demagnetised[8]. For rocks to acquire any magnetisation they must cool either during a prolonged period of constant polarity, or in a period when one polarity is dominant. Thus in a slowly cooled terrain one would expect either that the rocks should carry no stable magnetisation, or that they should all be magnetised with the same polarity.

The results reported here are from an area about 20 km square near Itivdleq some 40 km south of Holsteinsborg in western Greenland. The area is just within the early Proterozoic, Nagssugtoqidian Mobile Belt, close to its gradational boundary[9] with the Archaean craton to the south. The terrain comprises granulite and amphibolite facies gneisses cut by dolerite dykes which vary from fresh to almost completely autometamorphosed but which are essentially undeformed. There is strong geological evidence[10] that the area was at metamorphic temperatures when the dykes were intruded.

Stable magnetisation was found in both dykes (18 sites) and gneisses (4 sites), and after suitable alternating field (a.f.) demagnetisation most sites showed an unusually low within-site scatter of directions. Fourteen sites had a precision parameter, k, of better than 300 (corresponding to a circle of 95% confidence with a radius α_{95} of less than 3.2°), and at no site was k less than 50 ($\alpha_{95}=7.9°$). Largely because of the accuracy with which the site poles are defined the effects of slow cooling can be recognised.

Figure 1 shows that 18 poles (15 dykes and 3 gneisses) are arranged in a linear trend approximately 10° wide and

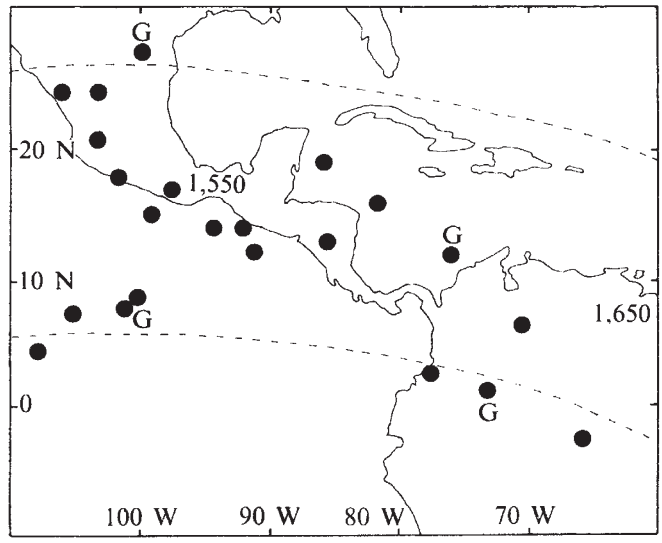


Fig. 1 The 22 site poles. Each is the mean of, generally, eight separately oriented specimens. The 18 dyke poles are unlabelled: G, the four gneiss poles. The poles have been rotated to correct for the drift of Greenland away from North America[11] to allow comparison with the appropriate part of the latest published APW path[12] for Laurentia (dashed band, with assigned ages in Myr). Most site poles initially showed southerly movements with progressive a.f. demagnetisation, presumably due to the removal of recently acquired viscous components, and at 7 sites (see text and Fig. 3) these were followed by systematic south-easterly movements. The pole selected for each site was generally either at the end point of the southerly movement or at the point where the southerly movement changed to a south-easterly movement, and was usually at a higher demagnetising field than that which gave the best within-site precision.

45° long, running NW–SE, and making a slight angle with the appropriate portion of the latest Laurentian APW path[12]. The remaining 4 poles are grouped together just to the south-west of the trend, which cannot be explained at present. All 174 specimens examined were magnetised with the same polarity. After correcting for the drift of Greenland away from North America[11] the mean direction of magnetisation for all 22 sites gives a pole at 13°N 91°W, with $k=47.1$ and α_{95} of 4.6°.

All of the sampling sites were within 50 m of sea level, so if the linear trend of poles is attributable to differences in the original structural level there must have been a slight tilting of the area subsequent to magnetisation. For example, a tilt of 5°, which would be difficult to detect geologically in such a terrain, would be equivalent over a horizontal distance of 20 km to a change in the original level of 1.8 km. This possibility is supported in this case by a general, though imperfect, relationship between the position of a pole in the linear trend of poles and the geographical position of the corresponding sampling site (Fig. 2): most of the sites in the NNW part of the area give poles in the south-eastern part of the trend, and sites in the SSE part of the area give poles in the north-western part of the trend. If the north-western part of the trend is the younger, the results suggest that the SSE part of the area represents the lowest structural level, that is, that there has been tilting of the terrain towards the NNW. The poles from sites D16, D17 and G20 (Fig. 2) clearly do not conform to this pattern; those sites are the most northerly in the area and so their poles should plot at the south-eastern and not the north-western end of the trend. But a prominent E–W fault, for which there is some geological evidence that the movement has been predominantly vertical (D. Nash, personal communication), separates them from the remainder of the sampling area. The apparent displacement of the anomalous poles could thus be caused by the area north of the fault having moved up with respect to the area to the south.

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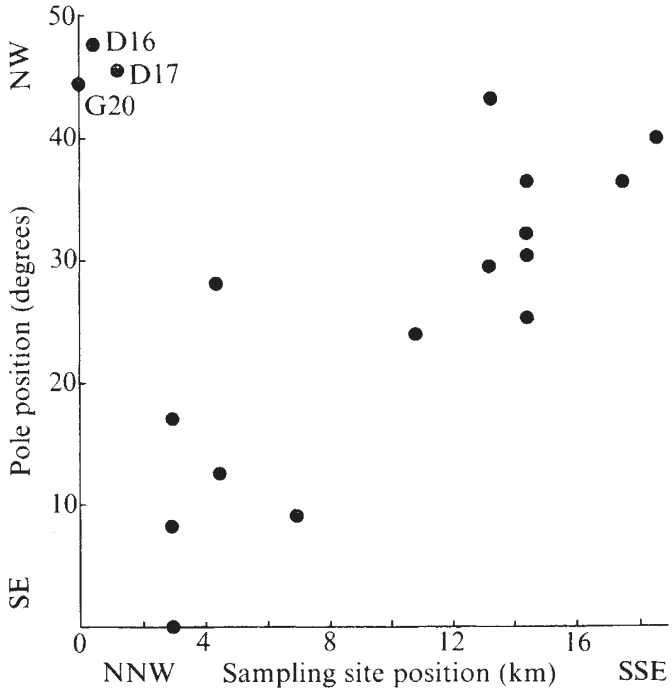


Fig. 2 The relationship between sampling site positions and corresponding pole positions for the 18 poles in the linear trend. Ordinate, the angular distance north-west along the linear trend to each pole from the D13 pole (the most south-easterly pole); abscissa, distance SSE to each sampling site from an imaginary ENE trending line through site G20 (the most northerly site).

If the linear trend of poles is due to APW during slow magnetisation then the scatter of poles attributable to other causes, represented by the 10° width of the trend, is equivalent to an angular standard deviation of about 4°. This is much less than the values of 9–24° quoted for secular variation for the appropriate palaeolatitude[13] indicating either that magnetisation was extremely fast, which seems inconceivable on geological grounds and would not in any case explain the linear trend, or that it was so slow that secular variation was effectively averaged out.

During progressive a.f. demagnetisation the site mean pole at seven sites showed small but very systematic movements always in a south-easterly direction (Fig. 3). As there is probably a reasonably close relationship between coercive force and blocking temperature, the systematic movements may represent the movement of the magnetisation from a younger direction, corresponding to grains with lower blocking temperatures, to an older one attributable to grains with higher blocking temperatures. Vector analysis of the movements indicates that, in general, the component actually removed in each step moves progressively in a south-easterly direction, suggesting either that there are two distinct magnetisations with overlapping coercivity spectra, or that there is a continuous spread of magnetisations. The latter seems more likely because the average direction of systematic movement is close to the trend of the poles, indicating that both record the APW path over that period of time. The linear trend of the poles is not greatly different in direction from the trend of the latest published APW path at that point[12], and the sense of time along the trend of poles as indicated by the systematic movements (younger towards the north-west and older towards the south-east) is in accord with the age dates assigned to the APW path.

Thermal demagnetisation studies were carried out on six specimens from sites which all showed systematic south-easterly movements during a.f. demagnetisation. Up to 520 °C the mean pole moved generally southwards but between 520 and 560 °C, when by far the major decrease in the intensity of magnetisation occurred, the mean pole moved systematically in a south-easterly direction very close to the mean direction of the systematic a.f. movements (Fig. 3). It therefore seems that the stable magnetisation of the specimens resides in grains of magnetite or titanomagnetite which have blocking temperatures (measured on the laboratory time scale) of between 520 and 560 °C, and that the systematic south-easterly movements also occur over that interval.

According to the assigned ages on the latest APW path the trend of poles covers an age range from about 1,525 to 1,650 Myr BP, but an examination of the well dated poles in the area suggests that this portion of the APW path may be about 100 Myr older than indicated. Ages are not yet available from the sampling area, but regional K–Ar mineral ages from the Nagssugtoqidian Belt as a whole range from 1,650 to 1,790 Myr BP, and provisional U–Pb studies on zircons give ages of 2,200–2,600 Myr BP (ref. 14). There thus seems to be good agreement between the time of magnetisation and the setting of the K–Ar mineral clocks, indicating that the magnetisation was probably acquired during a fairly late stage of uplift and cooling long after the peak of metamorphism.

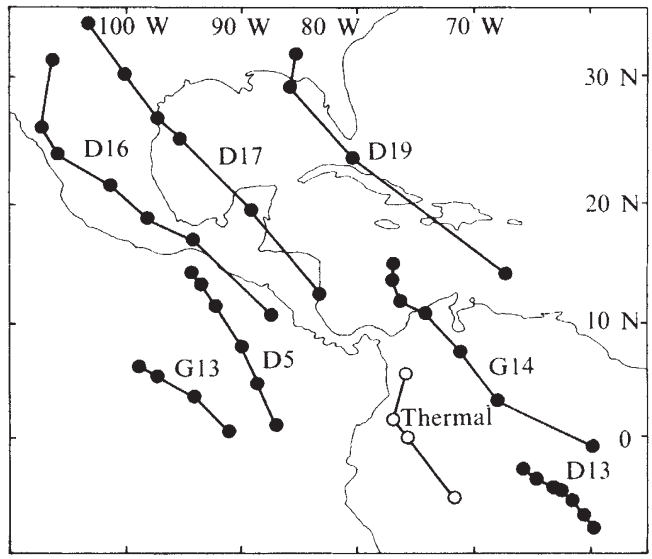


Fig. 3 Systematic south-easterly movements of the mean pole with progressive a.f. demagnetisation at seven sites. For several sites the final part of the southerly movement is also shown to illustrate the transition from southerly to south-easterly movement. ○, The movement of the mean pole of six selected specimens with progressive thermal demagnetisation between 500 and 560 °C. All of the poles have been corrected for the drift of Greenland as in Fig. 1. The poles for D17 and D19 have also been moved 10° north, and the thermal poles 9° south-west, to improve the clarity of the diagram. The demagnetising fields (oersted) were—D5: 200, 300, 400, 600, 800, 1,000; D13: 150, 200, 300, 400, 600, 800, 1,000; D16: 0, 50, 100, 200, 300, 400, 600; D17: 50, 100, 150, 200, 400, 600; D19: 100, 150, 200, 400; G13: 600, 800, 1,000, 1,200; G14: 100, 150, 200, 400, 600, 800, 1,000. For thermal demagnetisation the temperatures were 500, 520, 540 and 560 °C.

The results thus seem to show all of the characteristics to be expected from a slowly cooled terrain. If results of this type prove to be common in plutonic terrains they will not only provide valuable details about apparent polar wander, but could also be a useful tool for investigating movements, such as tilting and faulting, which have occurred subsequent to magnetisation, and which often cannot easily be studied by other methods.

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Effect of depth of burial and tectonic activity on coalification

THE study of variations in coal rank is important to coal geologists and the coal industry, and a proper understanding of the coalification process is also important to oil exploration geologists. The rank of coals, which occur as thin seams or lenses in oilfield sediments, can be regarded as an indicator for the degree of maturation of the organic matter (kerogen), finely dispersed in the sediments. Coal and kerogen are somewhat similar in chemical composition, and so are controlled by the same physical and chemical factors. If tectonic pressure can be shown to advance coalification, then it follows that a higher maturation of kerogen can be expected near a fault zone.

Coalification occurs when original plant material becomes buried, and gradual changes in its chemical composition and physical properties take place. The changes can lead to the formation of lignite, brown coal, bituminous coal or anthracite. During the late nineteenth century, it was widely supposed that burial pressure on coal was responsible for the advancement of coalification. In some tectonically disturbed coalfields, for instance, the presence of high rank coals were found which led geologists to believe that tectonic pressures accelerated coalification[1]. Other studies in coalfield geology and coal chemistry have convincingly shown that the role of pressure is insignificant in the process of coalification[2–6], and the results of experiments simulating coalification[6] indicate that the lower the pressure, the more rapid the coalification at a given temperature. From thermodynamics it is known that the gross reaction in chemical processes, including coalification, is determined mainly by temperature and time, and that static pressure retards reactions because it restricts the removal of reaction products[3]. A coalification pattern which does not seem to fit these studies has been reported for the Bowen Basin of Queensland, Australia (see ref. 7; and Fig. 1). We believe, however, that the pattern in this area is attributable to differences in the original depths of burial below subsequently eroded sediments.

During the Palaeozoic, a sediment-filled, linear downwarp formed along the eastern margin of Australia, separating an old land mass to the west from a rising orogenic belt to the east. During the Permian, coal-bearing formations were deposited; they were followed by as much as 5 km

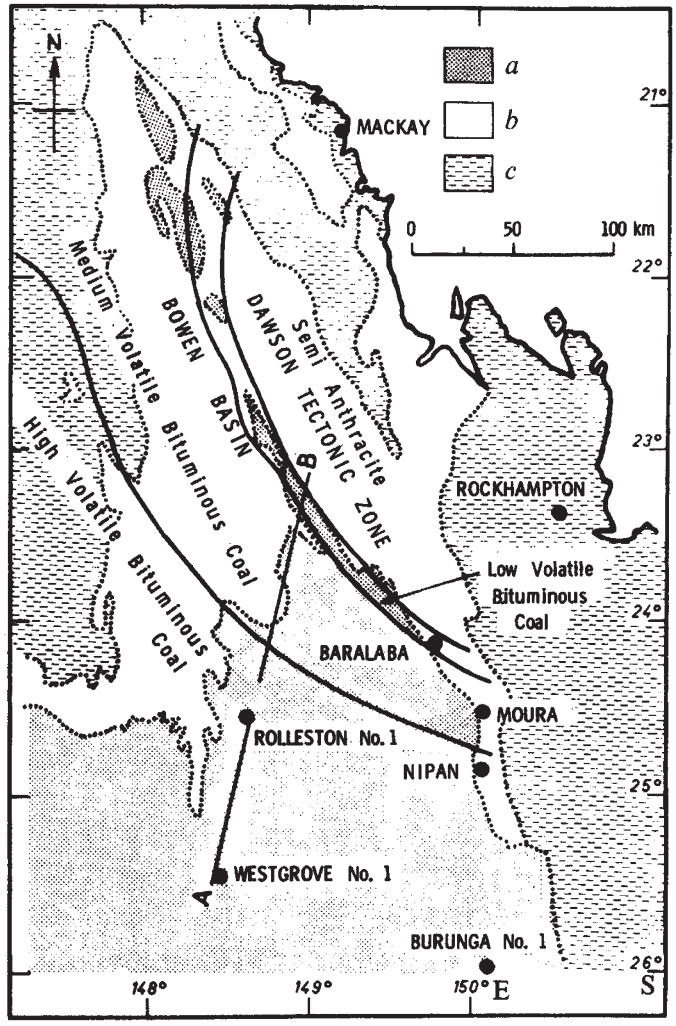


Fig. 1 Coalification trends in the Bowen Basin (after ref. 7) A–B, Location of cross section shown in Fig. 2. *a*, Mesozoic sediments; *b*, Permian sediments; *c*, pre-Permian sediments.

of Triassic sediments. After the deposition of these sediments, the area, particularly the northern part, became uplifted and suffered intense erosion. Contemporaneously, a large synclinal structure formed along the downwarp, the northern part of which is now the Bowen Basin. Tectonic activity was particularly strong along the eastern flank of the Bowen Basin, and the so-called Dawson Tectonic Zone formed. In areas close to this zone, semi-anthracite coals are now exposed at the surface. With increasing distance from this zone towards the south-west, the rank of outcropping coal seems to decrease regularly, down to that of high volatile bituminous coals over a distance of 100 km. This coalification pattern may have been generated by the formation of the Dawson Tectonic Zone (see, for example, ref. 7). Possibly, the tectonic activity together with the deep burial of the coal seams may have been responsible for the increase in rank[8], though pressure tends to inhibit rather than promote increase in rank[9]. Shibaoka *et al*[10], however, have shown that temperature increases attributable to deep burial were probably responsible for the increase in rank around the Dawson Tectonic Zone.

No evidence has been given to explain why the degree of tectonic disturbance gradually and regularly decreases away from the Dawson Tectonic Zone—only the zone itself is heavily disturbed. Even if tectonic pressure did decrease regularly over 200 km, it would not affect the coalification over such a distance[2–6]. The only real effect that the

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